

```
In [1]: import pandas as pd
```

```
In [2]: df = pd.read_csv("custdata.tsv", sep="\t")
```

```
In [3]: df.head()
```

```
Out[3]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th> <th>custid</th>
<th>sex</th> <th>is.employed</th>
<th>income</th> <th>marital.stat</th>
<th>health.ins</th> <th>housing.type</th>
<th>recent.move</th> <th>num.vehicles</th>
<th>age</th> <th>state.of.res</th> </thead>
<tbody> <th>0</th> <th>1</th> <th>2</th>
<th>3</th> <th>4</th> </tbody> <tbody></tbody>
```

2068	F	NaN	11300	Married	True	Homeowner free and clear
------	---	-----	-------	---------	------	--------------------------------

2073	F	NaN	0	Married	True	Rented
------	---	-----	---	---------	------	--------

2848	M	True	4500	Never Married	False	Rented
------	---	------	------	------------------	-------	--------

5641	M	True	20000	Never Married	False	Occupied with no rent
------	---	------	-------	------------------	-------	--------------------------

6369	F	True	12000	Never Married	True	Rented
------	---	------	-------	------------------	------	--------

```
In [4]: df[["age", "income", "num.vehicles"]].head
```

```
Out[4]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th> <th>age</th>
<th>income</th> <th>num.vehicles</th> </thead>
<tbody> <th>0</th> <th>1</th> <th>2</th>
<th>3</th> <th>4</th> </tbody> <tbody></tbody>
```

49.0	11300	2.0
------	-------	-----

40.0	0	3.0
------	---	-----

22.0	4500	3.0
------	------	-----

22.0	20000	0.0
------	-------	-----

31.0	12000	1.0
------	-------	-----

```
In [5]: before_corr = df[["age", "income", "num.ve
before_corr
```

```
Out[5]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th> <th>age</th>
<th>income</th> <th>num.vehicles</th> </thead>
<tbody> <th>age</th> <th>income</th>
<th>num.vehicles</th> </tbody> <tbody></tbody>
```

1.000000	0.027429	-0.063666
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0.027429	1.000000	0.139198
----------	----------	----------

-0.063666	0.139198	1.000000
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```
In [6]: df["age"].describe()
```

```
Out[6]: count      1000.000000
        mean       51.699815
        std        18.863433
        min         0.000000
        25%        38.000000
        50%        50.000000
        75%        64.000000
        max       146.680197
        Name: age, dtype: float64
```

```
In [7]: df[df["age"] > 100][["age", "income", "num
```

```
Out[7]: <style scoped> .dataframe tbody tr th:only-of-type {
        vertical-align: middle; } .dataframe tbody tr th {
        vertical-align: top; } .dataframe thead th { text-align:
        right; } </style> <thead> <th></th> <th>age</th>
        <th>income</th> <th>num.vehicles</th> </thead>
        <tbody> <th>212</th> <th>266</th> <th>286</th>
        <th>593</th> <th>625</th> <th>752</th>
        <th>926</th> <th>953</th> </tbody> <tbody>
        </tbody>
```

137.700030	60000	2.0
123.061023	12000	1.0
136.052160	31200	2.0
146.680197	40000	1.0
124.845087	175200	3.0
130.401605	200000	4.0
126.737284	43400	1.0
113.337207	255000	1.0

```
In [8]: df[df["age"] < 18][["age", "income", "num.
```

```
Out[8]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th> <th>age</th>
<th>income</th> <th>num.vehicles</th> </thead>
<tbody> <th>667</th> <th>868</th> <th>965</th>
</tbody> <tbody></tbody>
```

0.0	42400	2.0
-----	-------	-----

0.0	48000	3.0
-----	-------	-----

0.0	113950	2.0
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```
In [9]: df[df["income"] < 0][["age", "income", "nu
```

```
Out[9]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th> <th>age</th>
<th>income</th> <th>num.vehicles</th> </thead>
<tbody> <th>691</th> </tbody> <tbody></tbody>
```

60.0	-8700	3.0
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```
In [10]: (df["income"] < 0).sum()
```

```
Out[10]: np.int64(1)
```

```
In [11]: clean_df = df[
    (df["age"] >= 18) &
    (df["age"] <= 100) &
    (df["income"] >= 0)
].copy()
```

```
In [12]: clean_df.shape
```

Out[12]: (988, 11)

```
<p>after_corr = clean_df[["age", "income",  
"num.vehicles"]].corr()</p> <p>after_corr</p>
```

<p>I removed 11 rows from the age rows, which is represent impossible age which is less than 18 and older than 100. in addition i removed 1 row from the revenue row cause it is represent a negative number which is impossible too.</p>

<p>After cleaning the data, relation between age and income had almost no linear correlation ($r = 0.009$). relation between Age and number of vehicles also had a very weak negative correlation ($r = -0.066$), while relation between income and number of vehicles had a weak positive correlation ($r = 0.138$). Overall, the cleaned data does not show strong linear relationships among age, income, and number of vehicles.</p>